

LAB 3 [Group # → 1 to 8]

[Platform → HACKERRANK]

Introduction To Programming

(28-01-22)

SOLUTIONS

Q1 Vaccines and Recoveries

```
def numVaccine(n):  
    v1 = 0  
    v2 = 1  
    v3 = 1  
  
    if n==1:  
        return 0  
    if n==2:  
        return 1  
    if n==3:  
        return 1  
    else:  
        vacc_sum=0  
        for x in range(4, n+1):  
            vacc_sum = v1+v2+v3  
            v1=v2  
            v2=v3  
            v3=vacc_sum  
        return vacc_sum  
  
def numRecovery(n):  
    r1 = 0  
    r2 = 1  
    if n==1:  
        return 0  
    if n==2:  
        return 1  
    else:  
        rec_sum=0  
        for x in range(3, n+1):  
            rec_sum = r1+r2  
            r1=r2  
            r2=rec_sum  
        return rec_sum  
  
n = int(input())  
diff = numVaccine(n) - numRecovery(n)  
print(diff)
```

Q2 Seconds passed from midnight

```

hr = int(input())
mn = int(input())

def is_valid(hr, mn):
    valid = 'Valid'
    if hr>24:
        valid = 'Invalid'
    if mn>60:
        valid = 'Invalid'

    return valid

t = is_valid(hr, mn)

if t=='Invalid':
    print(t)
else:
    ts = (hr*60+mn)*60
    print(ts)

```

Q3 Parity Analysis

```

def sumDigit(n):
    s = 0

    while(n!=0):
        s += n%10
        n =n//10

    return s

def parity(x):
    if x%2==0:
        return 'e'
    else:
        return 'o'

n = int(input())
if parity(n)== parity(sumDigit(n)):
    print('True')
else:
    print('False')

```

Q4 Interning at Staark Tower

```

def go():
    ll = input().split(" ")
    n = int(ll[0])
    x = int(ll[1])
    lst = []
    for i in range(n):
        lst.append(int(input()))
    for i in range(n):
        sm = 0;
        for j in range(i,n):
            sm += lst[j]
            d = j-i+1;
            if(sm == d*x):
                print("YES")
                return
    print("NO")
go()

```

Q5. Got A Chance

```

n = int(input())

halt = False
i = 0
while(not halt):
    if 2**i>=n:
        halt = True
        break

    i=i+1

if 2**i == n:
    print(0)
elif n==0:
    print(1)
else:
    if 2**i>n:
        d1 = n-2**(i-1)
        d2 = 2**i - n
        if d1<d2:
            print (d1)
        else:
            print(d2)

```

Q6. It's the salary day

```
basic_salary = float(input())

if basic_salary <= 10000:
    hra = 0.09 * basic_salary
    da = 0.11 * basic_salary

elif (10000 <= basic_salary) and (basic_salary <=20000):
    hra = 0.18 * basic_salary
    da = 0.22 * basic_salary
else:
    hra = 0.27 * basic_salary
    da = 0.33 * basic_salary

gross_salary = basic_salary + hra + da

print(f"{gross_salary:.2f}")
```

Q7 Population Growth

```
def time_taken(a0,p,n,t):
    a = a0
    time = 0
    while t > a:
        a += (p/100)*a + n
        time += 1
    return(time)

a0 = int(input())
p = float(input())
n = int(input())
t = int(input())
print(time_taken(a0,p,n,t))
```